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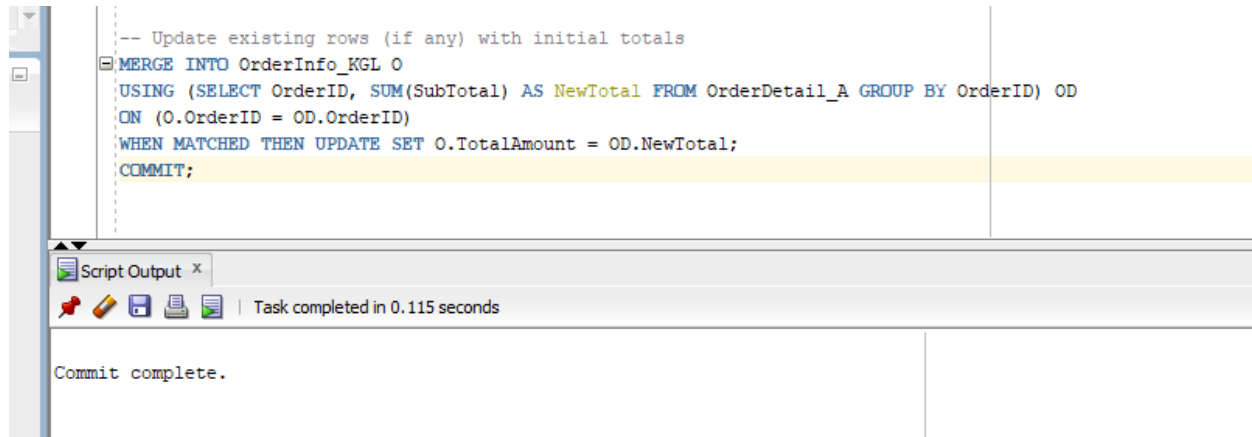
UR/ACE-DS

Masters of Data Science in Mining

Module: ADVANCED DATABASE TECHNOLOGY

FINAL EXAM

SECTION B EXAM SCREENSHOTS



The screenshot displays a database management interface. The top pane shows a SQL script for updating order totals. The script includes a comment, a MERGE statement, a subquery for calculating new totals, and a COMMIT. The bottom pane, titled 'Script Output', shows the message 'Commit complete.' and a status bar indicating the task was completed in 0.115 seconds.

```
-- Update existing rows (if any) with initial totals
MERGE INTO OrderInfo_KGL O
USING (SELECT OrderID, SUM(SubTotal) AS NewTotal FROM OrderDetail_A GROUP BY OrderID) OD
ON (O.OrderID = OD.OrderID)
WHEN MATCHED THEN UPDATE SET O.TotalAmount = OD.NewTotal;
COMMIT;
```

Script Output x

Task completed in 0.115 seconds

Commit complete.

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```
-- Failing INSERT 1 (Quantity = 0)
BEGIN
  INSERT INTO OrderDetail_A (OrderID, ProductID, Quantity, SubTotal) VALUES (999, 4, 0, 5.00);
EXCEPTION
  WHEN OTHERS THEN
    IF SQLCODE = -2290 THEN -- ORA-02290: check constraint violated
      DBMS_OUTPUT.PUT_LINE('✓ Fail Case 1: Quantity <= 0 failed with ORA-02290.');
```

ELSE
RAISE;
END IF;
END;

Script Output x

Task completed in 0.048 seconds

ERROR at line 7:
ORA-06550: line 7, column 17:
PLS-00201: identifier 'V_FAIL_COUNT' must be declared
ORA-06550: line 7, column 17:
PL/SQL: Statement ignored

<https://docs.oracle.com/error-help/db/ora-06550/>

More Details :
<https://docs.oracle.com/error-help/db/ora-06550/>
<https://docs.oracle.com/error-help/db/pls-00201/>

```
-- 3. SELECT proof that only the passing rows were committed
SELECT OrderID, Quantity, SubTotal FROM OrderDetail_A WHERE OrderID IN (106, 107);
SELECT COUNT(*) FROM OrderDetail_A; -- Should be <= 10
```

Query Result x

SQL | All Rows Fetched: 1 in 0.005 seconds

COUNT(*)
10

UPDATE OrderInfo_KGL SET TotalAmount = V_New_Total WHERE OrderID = V_OrderID_Key;

Script Output x

Task completed in 0.108 seconds

Trigger TRG_ORDERDETAIL_TOTAL compiled

Messages - Log

Sequence...	Connection Name	Elapsed	SQL	Parameters
807	S_BranchDB_A	4	SELECT LINE, POSITION, REPLACE(TEXT, CHR(10), '') attribute FROM SYS.ALL_ERRORS A ...	"OWNER"="S_BranchDB_A", "TYPE"="TRIGGER", "NAME"="TRG_ORI
806	S_BranchDB_A	1	select UPPER(sys_context(USERENV, 'CURRENT_SCHEMA')) from dual	
805	S_BranchDB_A	1	declare l_line varchar2(32767); l_done number; l_buffer varchar2(32767) := ""; l_lengthbuffer ...	1=1000
804	S_BranchDB_A	84	CREATE OR REPLACE TRIGGER trg_OrderDetail_TotalAFTER INSERT OR UPDATE OR DELETE ON Ord...	
803	S_BranchDB_A	3	SELECT LINE, POSITION, REPLACE(TEXT, CHR(10), '') attribute FROM SYS.ALL_ERRORS A ...	"OWNER"="S_BranchDB_A", "TYPE"="TRIGGER", "NAME"="TRG_ORI
802	S_BranchDB_A	1	select UPPER(sys_context(USERENV, 'CURRENT_SCHEMA')) from dual	
801	S_BranchDB_A	1	declare l_line varchar2(32767); l_done number; l_buffer varchar2(32767) := ""; l_lengthbuffer ...	1=1000

Messages | Statements x

```
-- 1. Create audit table
CREATE TABLE OrderInfo_AUDIT (
    OrderID NUMBER(10),
    Bef_Total NUMBER(12, 2),
    Aft_Total NUMBER(12, 2),
    Changed_At TIMESTAMP DEFAULT SYSTIMESTAMP
);
```

Script Output x

Task completed in 0.166 seconds

Table ORDERINFO_AUDIT created.

```
-- 2. Implement a statement-level AFTER DML trigger on OrderDetail_A

CREATE OR REPLACE TRIGGER trg_OrderDetail_Total
AFTER INSERT OR UPDATE OR DELETE ON OrderDetail_A
DECLARE
    V_OrderID_Key NUMBER(10);
    V_Old_Total NUMBER(12, 2);
    V_New_Total NUMBER(12, 2);
BEGIN
    -- This block executes once per statement (Statement-level)
    -- Update all affected OrderInfo rows (simpler logic than row-level for small scale)
    FOR rec IN (
        SELECT DISTINCT OrderID FROM OrderDetail_A
        -- In a real scenario, we'd check OLD/NEW table for affected OrderIDs
    ) LOOP
        V_OrderID_Key := rec.OrderID;

        -- Get old total before re-computation
        SELECT TotalAmount INTO V_Old_Total FROM OrderInfo_KGL WHERE OrderID = V_OrderID_Key;

        -- Re-compute the total for the OrderID
        SELECT NVL(SUM(SubTotal), 0) INTO V_New_Total FROM OrderDetail_A WHERE OrderID = V_OrderID_Key;

        -- Update the denormalized total
        UPDATE OrderInfo_KGL SET TotalAmount = V_New_Total WHERE OrderID = V_OrderID_Key;

        -- Log to audit table
        INSERT INTO OrderInfo_AUDIT (OrderID, Bef_Total, Aft_Total)
        VALUES (V_OrderID_Key, V_Old_Total, V_New_Total);
    END LOOP;
END;
```

```
-- 4. Audit validation
SELECT OrderID, TotalAmount FROM OrderInfo_KGL WHERE OrderID IN (106, 107);
-- Expected: OrderID 106 TotalAmount = 6.00 (from 5.00)
-- Expected: OrderID 107 TotalAmount = 10.00 (from 12.00+10.00 - 12.00 = 10.00 total)

-- Expected Audit entries (2-3 entries, one for each statement)
SELECT * FROM OrderInfo_AUDIT ORDER BY Changed_At DESC;
```

Query Result x

SQL | All Rows Fetched: 0 in 0.016 seconds

ORDERID	BEF_TOTAL	AFT_TOTAL	CHANGED...
---------	-----------	-----------	------------

```
-- 3. Execute a small mixed DML script (Reuse existing OrderID 106, 107, and others)
-- We need to ensure OrderID 106 and 107 exist in OrderInfo_KGL for the trigger to find V_Old_Total
INSERT INTO OrderInfo_KGL (OrderID, CustomerID, OrderDate, TotalAmount) VALUES (106, 12, DATE '2025-10-28', 5.00);
INSERT INTO OrderInfo_KGL (OrderID, CustomerID, OrderDate, TotalAmount) VALUES (107, 13, DATE '2025-10-28', 12.00);
COMMIT; -- Ensure base data exists
```

Script Output x

Task completed in 0.155 seconds

Commit complete.

```
SET SERVEROUTPUT ON
DECLARE
  -- These are test rows that we will ROLLBACK
  V_FAIL_COUNT NUMBER := 0;
BEGIN
  -- Passing INSERT 1 (OrderID=106, will be committed)
  INSERT INTO OrderDetail_A (OrderID, ProductID, Quantity, SubTotal) VALUES (106, 4, 1, 5.00);
```

Script Output x

Task completed in 0.05 seconds

ORA-06550: line 9, column 1:
PLS-00103: Encountered the symbol "end-of-file" when expecting one of the following:

(begin case declare end exception exit for goto if loop mod
null pragma raise return select update while with
<an identifier> <a double-quoted delimited-identifier>
<a bind variable> << continue close current delete fetch lock
insert open rollback savepoint set sql execute commit forall
merge standard pipe purge json_object

<https://docs.oracle.com/error-help/db/ora-06550/>

```
-- DML Script
UPDATE OrderDetail_A SET SubTotal = 6.00 WHERE OrderID = 106 AND ProductID = 4; -- UPDATE 1: Order 106 total changes
INSERT INTO OrderDetail_A (OrderID, ProductID, Quantity, SubTotal) VALUES (107, 6, 1, 10.00); -- INSERT 1: Order 107 total changes
DELETE FROM OrderDetail_A WHERE OrderID = 107 AND ProductID = 5; -- DELETE 1: Order 107 total changes (original row)
COMMIT;
```

Script Output x

Task completed in 0.129 seconds

Commit complete.

```
-- 1. Add NOT NULL and CHECK constraints on OrderDetail_B
ALTER TABLE OrderDetail_B ADD CONSTRAINT NN_ODB_QTY CHECK (Quantity IS NOT NULL) ENABLE NOVALIDATE;
ALTER TABLE OrderDetail_B ADD CONSTRAINT CK_ODB_QTY CHECK (Quantity > 0) ENABLE NOVALIDATE;
ALTER TABLE OrderDetail_B ADD CONSTRAINT CK_ODB_SUBTOTAL CHECK (SubTotal > 0) ENABLE NOVALIDATE;
```

Script Output x

Task completed in 0.105 seconds

Table ORDERDETAIL_B altered.

Table ORDERDETAIL_B altered.

```
-- SECTION B
-- 1. Add NOT NULL and CHECK constraints on OrderDetail_A
ALTER TABLE OrderDetail_A ADD CONSTRAINT NN_ODA_QTY CHECK (Quantity IS NOT NULL) ENABLE NOVALIDATE;
ALTER TABLE OrderDetail_A ADD CONSTRAINT CK_ODA_QTY CHECK (Quantity > 0) ENABLE NOVALIDATE;
ALTER TABLE OrderDetail_A ADD CONSTRAINT CK_ODA_SUBTOTAL CHECK (SubTotal > 0) ENABLE NOVALIDATE;
```

Script Output x

Task completed in 0.102 seconds

Table ORDERDETAIL_A altered.